IncomeIQ

Smart Income Prediction Using Machine Learning

5th Semester AIML Mini Project Report

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1. INTRODUCTION

IncomeIQ is a machine learning project that predicts whether an individual's income is above $50K using demographic and employment-related attributes. The task is framed as a **binary classification** problem and is solved using the **Random Forest Classifier**, an ensemble learning algorithm that combines multiple decision trees to produce a robust prediction.

2. DATASET

The project uses the **Adult Census Income Dataset**. After cleaning, the dataset contains **32,537 records**, and categorical encoding produced **100 encoded features**. Source: Kaggle – Adult Income Classification

3. DATA PREPROCESSING AND METHODOLOGY

1. **Loading the dataset:** The Adult Census Income dataset was loaded for analysis.
2. **Data exploration:** The dataset structure and attributes were examined.
3. **Handling missing values:** Missing values in the data were identified and handled.
4. **Removing missing-value markers:** Rows containing missing-value markers were removed, leaving 32,537 records.
5. **One-Hot Encoding:** Categorical variables were converted into numerical form, resulting in 100 encoded features.
6. **Train-test split:** The data was divided into training and testing sets in an 80:20 ratio.
7. **Model training and prediction:** The Random Forest Classifier was trained on the training set and used to predict income on the test set.

4. MODEL EVALUATION RESULTS

Metric	Value
Accuracy	**84.96%**
Precision (income above $50K)	**72.36%**
Recall (income above $50K)	**60.78%**
F1-Score (income above $50K)	**66.07%**

5. KEY FINDINGS

The Random Forest model achieved an accuracy of **84.96%** on the test data. Feature importance analysis was used to understand which input features influence the income predictions made by the model.

6. CONCLUSION

This project demonstrates that the Random Forest algorithm can be applied to real-world income classification tasks, where individuals must be categorised into income groups based on their demographic and employment attributes. Through the project, the importance of careful data preprocessing, including handling missing values and one-hot encoding of categorical variables, was understood as a foundation for effective model training. Evaluating the model with accuracy, precision, recall and F1-score gave a balanced view of its performance, while feature importance analysis provided insight into which features drive the model's predictions, making the results more interpretable.